Lingchen Meng

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RESEARCH INTERESTS

Systems Security: Cyber-Physical Systems Security; Computer System Security
AI/Robotics Security: Embodied AI Security

EDUCATION

Ph.D. in Computer Science **Aug 2025 – Present**
Advisor: Dr. Ning Zhang
Washington University in St. Louis, St. Louis, MO

B.S. in Computer Science **Aug 2022 – May 2025**
Washington University in St. Louis, St. Louis, MO
GPA: 3.98/4.00
Honors: Summa Cum Laude

HONORS

- Dean's Select PhD Fellowship
- Tau Beta Pi Engineering Honor Society Member
- Dean's List, Fall 2022 – Spring 2025

Ongoing Research

Physical Adversarial Examples in ML Perception Systems
Researcher *Fall 2025 - Present*

- Modeled physical adversarial examples as an end-to-end physical-to-digital attack pipeline
- Surveyed 130+ prior works and analyzed limitations in existing robustness evaluations
- Built a controlled experimental platform to evaluate how physical and sensing factors affect adversarial attack reliability

Measuring Impact Hardness of AI Command Errors in Embodied Control Stacks
Researcher *Spring 2026 - Present*

- Perform static analysis on AI-enabled control stack, such as self-driving and humanoid.
- Investigate the VLA model and the control logic of the Unitree G1 Humanoid

Work Experience

AR Admission Letter of The High School Affiliated to Renmin University of China **Beijing, China**
Leader, Developer *Summer 2023, Winter 2023*

- Held weekly meetings to check in with teammates, assign work and requirements
- Established AR environment and designed interactive elements
- Managed the application distribution for the iOS system and kept track of user data

Introduction to Computer Engineering & Introduction to Computer Security **Washington University**
Teaching Assistant *Fall 2023 - Fall 2024*

- Held in-person office hours and answered students' questions
- Helped students with their in-class studio
- Graded assignments and exams

Technical Project

Electromagnetic Interference Ledger & Registry (EMILY) **Washington University**
Project Developer *Fall 2025*

- Designed a real-time EMI event registry that allows users to upload identified electromagnetic interference events and visualize them on a global map
- Implemented fine-grained role-based access control for individual users, organizations, and super administrators to manage event submission, verification, and visibility

Reverse Engineering Camera Slider APK **Washington University**
Project Developer *Summer 2024*

- Used both static and dynamic analyses to gather information on APK file
- Based on the behavior, reversed the BLE protocol and designed script to precisely control slider

Reverse Engineering Minesweeper Game **Washington University**
Project Developer *Spring 2024*

- Used both static and dynamic analyses to gather information on exe file
- Analyzed assembly code from the executable
- Modified assembly code to customize finish time, stop the clock at a particular time, and show the position of bombs immediately after the game starts

Memory Allocation Simulation **Washington University**
Project Developer *Fall 2023*

- Implemented a heap simulator supporting heap initialization, allocation, free, reallocation, contiguous allocation, extend heap, and coalesce.
- Used explicit free list to increase throughput and N-th fit to increase space utilization.

Simple Calendar Website Development **Washington University**
Project Developer *Fall 2023*

- Developed a user registration and login system and stored the usernames and hashed and salted passwords in SQL database
- Implemented functionalities for users to add, delete, and edit events, which are displayed on the calendar with different colors based on the event category
- Website reminds users 5 minutes before the starting time of an event

Simple Chat Room Website Development **Washington University**
Project Developer *Fall 2023*

- Developed functionalities for users to create public or private rooms with password protection
- Implemented chat functionalities for public and private messages within the room
- Administrator can assign or remove co-admin, mute or unmute, kick, and ban users.

Communication with AES-GCM Encryption Project **Washington University**
Project Developer *Fall 2023*

- Developed a messaging system based on socket between client and server with AES-GCM encryption
- Ensured both confidentiality and integrity with this encryption

File Managing System Development **Washington University**
Project Developer *Spring 2023*

- Developed a file management system similar to a Linux
- Supported commands: cat, copy, display, ls, remove, rename, and touch
- Implemented password protection for files

Skills

Programming Skills:
C/C++, Python, Java, JavaScript, HTML, CSS, PHP, SQL, R, MATLAB, Assembly, LateX
Software:
Visual Studio, VS Code, AWS EC2, MATLAB, gdb, Wireshark, IDA Pro, Immunity Debugger, PE-View, Metasploit
Languages:
English, Chinese